MS2 Summary Swissgrid Control Area Balance

What I achieved during Milestone 2

1. Fixed the forecast issue time in code: 23:20 UTC, all 96 steps of D+1, non-recursive (transform.py).
2. Built the feature pipeline from a “what is known when”. Rule: only lags of 192 and 672 steps (the t-96 lag is not available at 23:20 UTC). The test tests/test_transform.py checks that no feature uses a value published after the issue time.
3. Recomputed the persistence baseline on the out-of-time split (Test starts 01.07.2026, 11,187 rows). It was 103.105 MW with the unavailable one-day lag. It is now 96.09 MW (t-192) and 94.26 MW (t-672) (baseline.py).
4. Containerised the ingestion and set up GitHub Actions (daily_ingest_batch.yml) with a daily run that tests ingestion. Raw files are saved with dates.
5. Cost / honest limits:
    a. The 2024 and 2025 files are in Zurich local time with different column names, so the transform needed a rewrite to unify them (tests cover both). All sql tables are in UTC, the final staging table will be converted back into Zürich local time.
    b. The price feature only starts 01.01.2025, which leaves about 62.9k rows with all features.
    c. There is no persistent storage yet. GCS is planned for MS3, so the CI data is discarded after each run.
6. Look first at: transform.py, tests/test_transform.py, baseline.py, daily-batch.yml, README “What is known when”.

Review Response

A1 / A2 · Features not available at forecast time - Changed (e14f652)
Issue time fixed at 23:20 UTC. Lags t-1 and t-4 removed. The price (AE-Preis) is used only lagged by at least 192 steps, so it cannot leak the label. Rule documented in README “What is known when”. Evidence: transform.py, tests/test_transform.py.

B2 · First-published vs corrected values – Partly (88dfcf8)
Raw downloads are kept with dates and never overwritten, and daily snapshots daily-YYYY-MM-DD exist. The live MAE will be scored against first-published values (the daily snapshot). A second figure against corrected values follows after the revision window. Not done: persistent storage (GCS, MS3), so the history is not kept between CI runs yet. The forecast table with creation time and model version comes in M3.

B1 · Re-predicting vs evaluating - Partly
The policy is now written in the proposal: the original forecast is the evaluated one, and recalculations are labelled separately. Not done: the flag in the forecast table (M3).

E1 · Volume, access, fallback - Changed (88dfcf8)

The README Data section now has the volume (69,463 rows), the access method, the terms of use and ENTSO-E as fallback, however, ENTSO-E has not been implemented in M2 and will only be set up should Swissgrid stop delivering the required data.

E2 · Lighter core - Partly
The core is now GitHub Actions cron (daily_ingest_batch.yml). Parquet on GCS and hosted MLflow come with the GCS move in MS3. Cloud SQL is marked optional in the proposal.

E3 · Diagram and text fixes - Changed
Fig. 1 update and text fixes follow with the proposal changes (data source, registry, monitoring added; “DSPRO1/2” removed; “trepidation” changed to “precipitation”).

Points no reviewer asked: weather/precipitation is not ingested in MS2 since it is labelled optional. If time permits it is planned for M3. Due to reduced data the 3 to 7 day horizon is out of scope (lag 192 does not cover D+2).